RIVERINE FLOODS AND URBAN FLOOD RESILIENCE

FLOOD-FAMILY MATRIX → CATCHMENT-TO-CITY CHAIN → URBAN RUNOFF MECHANISM → WATER-SPACE FIREWALL → COMPOUND COASTAL FLOOD MAP → CWC FORECAST RAIL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◆ SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artifacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

00

STAGE 00 FLOOD-FAMILY MATRIX

FLOOD-FAMILY MATRIX · FLOODPLAIN OVERFLOW · RAINFALL EXCEEDS LOCAL STORAGE · RAPID CONCENTRATED RUNOFF OR SUDDEN RELEASE · BUILT CATCHMENT OR SEA INTERACTION

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
RIVERINE → channel / floodplain overflow	FLASH → rapid concentrated runoff or sudden release	RIVERINE → channel / floodplain overflow
PLUVIAL → rainfall exceeds local storage / drainage	URBAN / COASTAL → built catchment or sea interaction	PLUVIAL → rainfall exceeds local storage / drainage

HAZARD / ACTOR / INSTRUMENT
- RIVERINE → channel / floodplain overflow
- PLUVIAL → rainfall exceeds local storage / drainage

DECISIVE RISK OR MANDATE DISTINCTION
- FLASH → rapid concentrated runoff or sudden release
- URBAN / COASTAL → built catchment or sea interaction

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- RIVERINE → channel / floodplain overflow
- PLUVIAL → rainfall exceeds local storage / drainage

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: URBAN / COASTAL → built catchment or sea interaction.

01

STAGE 01 CATCHMENT-TO-CITY CHAIN

CATCHMENT-TO-CITY CHAIN · RAINFALL + ANTECEDENT MOISTURE + SLOPE · LAND COVER · TRIBUTARY TIMING + CHANNEL · SEDIMENT CONDITION · CITY EXPOSURE · DRAINAGE LAND USE WARNING AND CAPACITY

RAINFALL + ANTECEDENT MOISTURE + SLOPE / LAND COVER → TRIBUTARY TIMING + CHANNEL / SEDIMENT CONDITION → FLOODPLAIN / CITY EXPOSURE → OUTCOME → drainage land use warning and capacity

RISK CONDITION / INPUT
- RAINFALL + ANTECEDENT MOISTURE + SLOPE / LAND COVER
- TRIBUTARY TIMING + CHANNEL / SEDIMENT CONDITION

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- FLOODPLAIN / CITY EXPOSURE
- OUTCOME → drainage land use warning and capacity

LAST-MILE / INCLUSION / OUTCOME LIMIT
- RAINFALL + ANTECEDENT MOISTURE + SLOPE / LAND COVER
- TRIBUTARY TIMING + CHANNEL / SEDIMENT CONDITION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → drainage land use warning and capacity.

02

STAGE 02 URBAN RUNOFF MECHANISM

URBAN RUNOFF MECHANISM · IMPERVIOUS SURFACE · FASTER RUNOFF · SYNCHRONISED PEAK · OVERBURDENED DRAIN · DENSE EXPOSURE · SERVICE AND LIVELIHOOD LOSS

IMPERVIOUS SURFACE → FASTER RUNOFF → SYNCHRONISED PEAK → OVERBURDENED DRAIN → BLOCKAGE / ENCROACHMENT → BACKFLOW / WATERLOGGING → DENSE EXPOSURE → SERVICE AND LIVELIHOOD LOSS

RISK CONDITION / INPUT
- IMPERVIOUS SURFACE → FASTER RUNOFF
- SYNCHRONISED PEAK → OVERBURDENED DRAIN

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- BLOCKAGE / ENCROACHMENT → BACKFLOW / WATERLOGGING
- DENSE EXPOSURE → SERVICE AND LIVELIHOOD LOSS

LAST-MILE / INCLUSION / OUTCOME LIMIT
- IMPERVIOUS SURFACE → FASTER RUNOFF
- SYNCHRONISED PEAK → OVERBURDENED DRAIN

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BLOCKAGE / ENCROACHMENT → BACKFLOW / WATERLOGGING; DENSE EXPOSURE → SERVICE AND LIVELIHOOD LOSS.

03

STAGE 03 WATER-SPACE FIREWALL

WATER-SPACE FIREWALL · ROOM FOR RIVER EXPANSION · STORAGE AND ATTENUATION · NATURAL DRAIN · CONVEYANCE PATH · EXPOSURE + LOST HYDRAULIC FUNCTION

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
FLOODPLAIN → room for river expansion	NATURAL DRAIN → conveyance path	FLOODPLAIN → room for river expansion
WETLAND / LAKE → storage and attenuation	ENCROACHMENT → exposure + lost hydraulic function	WETLAND / LAKE → storage and attenuation

HAZARD / ACTOR / INSTRUMENT
- FLOODPLAIN → room for river expansion
- WETLAND / LAKE → storage and attenuation

DECISIVE RISK OR MANDATE DISTINCTION
- NATURAL DRAIN → conveyance path
- ENCROACHMENT → exposure + lost hydraulic function

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- FLOODPLAIN → room for river expansion
- WETLAND / LAKE → storage and attenuation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NATURAL DRAIN → conveyance path; ENCROACHMENT → exposure + lost hydraulic function.

04

STAGE 04 COMPOUND COASTAL FLOOD MAP

COMPOUND COASTAL FLOOD MAP · RIVER FLOW · PLUVIAL RUNOFF · URBAN DRAIN · DOWNSTREAM IMPEDANCE · COMPOUND INUNDATION

RIVER FLOW → ESTUARY

PLUVIAL RUNOFF → URBAN DRAIN → SURGE / TIDE / WAVE → DOWNSTREAM IMPEDANCE → COMBINATION → COMPOUND INUNDATION

HAZARD / RISK / RESILIENCE CONCEPT
- RIVER FLOW → ESTUARY
- PLUVIAL RUNOFF → URBAN DRAIN

INSTITUTIONAL OR COMMUNITY RESPONSE
- SURGE / TIDE / WAVE → DOWNSTREAM IMPEDANCE
- COMBINATION → COMPOUND INUNDATION

CAPACITY / STATUS / EVIDENCE LIMIT
- RIVER FLOW → ESTUARY
- PLUVIAL RUNOFF → URBAN DRAIN

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SURGE / TIDE / WAVE → DOWNSTREAM IMPEDANCE.

05

STAGE 05 CWC FORECAST RAIL

CWC FORECAST RAIL · 1 OBSERVE LEVEL · 2 ANALYSE AND FORECAST · 3 ISSUE SITE-SPECIFIC CATEGORY · 4 ADMINISTRATION · PROJECT AUTHORITY ACTS

1 OBSERVE LEVEL / DISCHARGE / INFLOW → 2 ANALYSE AND FORECAST → 3 ISSUE SITE-SPECIFIC CATEGORY / MESSAGE → 4 ADMINISTRATION / PROJECT AUTHORITY ACTS

HAZARD / RISK / RESILIENCE CONCEPT
- 1 OBSERVE LEVEL / DISCHARGE / INFLOW
- 2 ANALYSE AND FORECAST

INSTITUTIONAL OR COMMUNITY RESPONSE
- 3 ISSUE SITE-SPECIFIC CATEGORY / MESSAGE
- 4 ADMINISTRATION / PROJECT AUTHORITY ACTS

CAPACITY / STATUS / EVIDENCE LIMIT
- 1 OBSERVE LEVEL / DISCHARGE / INFLOW
- 2 ANALYSE AND FORECAST

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 3 ISSUE SITE-SPECIFIC CATEGORY / MESSAGE; 4 ADMINISTRATION / PROJECT AUTHORITY ACTS.

06

STAGE 06 RESERVOIR GOVERNANCE BOUNDARY

RESERVOIR GOVERNANCE BOUNDARY · INFLOW FORECAST + CURRENT STORAGE · OPERATING RULE · DAM CONDITION · DOWNSTREAM CAPACITY · GATE DECISION · TIMELY DOWNSTREAM WARNING · DAM IS NEITHER GUARANTEE NOR SINGLE CAUSE

INFLOW FORECAST + CURRENT STORAGE

OPERATING RULE / DAM CONDITION / DOWNSTREAM CAPACITY → GATE DECISION → TIMELY DOWNSTREAM WARNING → TRAP → DAM IS NEITHER GUARANTEE NOR SINGLE CAUSE

HAZARD / ACTOR / INSTRUMENT
- INFLOW FORECAST + CURRENT STORAGE
- OPERATING RULE / DAM CONDITION / DOWNSTREAM CAPACITY

DECISIVE RISK OR MANDATE DISTINCTION
- GATE DECISION → TIMELY DOWNSTREAM WARNING
- TRAP → DAM IS NEITHER GUARANTEE NOR SINGLE CAUSE

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- INFLOW FORECAST + CURRENT STORAGE
- OPERATING RULE / DAM CONDITION / DOWNSTREAM CAPACITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → DAM IS NEITHER GUARANTEE NOR SINGLE CAUSE.

07

STAGE 07 STRUCTURAL MEASURE TEST

STRUCTURAL MEASURE TEST · STORAGE AND OPERATION LIMITS · LOCAL PROTECTION AND BREACH · TRANSFER RISK · CONVEYANCE AND DOWNSTREAM EFFECTS · CAPACITY DEPENDS ON OUTLET AND MAINTENANCE

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
DAM / RESERVOIR → storage and operation limits	CHANNEL / DIVERSION → conveyance and downstream effects	DAM / RESERVOIR → storage and operation limits
EMBANKMENT → local protection and breach / transfer risk	DRAINAGE → capacity depends on outlet and maintenance	EMBANKMENT → local protection and breach / transfer risk

HAZARD / ACTOR / INSTRUMENT
- DAM / RESERVOIR → storage and operation limits
- EMBANKMENT → local protection and breach / transfer risk

DECISIVE RISK OR MANDATE DISTINCTION
- CHANNEL / DIVERSION → conveyance and downstream effects
- DRAINAGE → capacity depends on outlet and maintenance

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- DAM / RESERVOIR → storage and operation limits
- EMBANKMENT → local protection and breach / transfer risk

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DRAINAGE → capacity depends on outlet and maintenance.

08

STAGE 08 URBAN PLANNING PORTFOLIO

URBAN PLANNING PORTFOLIO · MASTER PLAN + FLOODPLAIN ZONING · DRAIN INVENTORY + CATCHMENT · CONTOUR DESIGN · SOLID WASTE + WATERWAY · WETLAND PROTECTION · UTILITY DEVELOPMENT CONTROL

MASTER PLAN + FLOODPLAIN ZONING

DRAIN INVENTORY + CATCHMENT / CONTOUR DESIGN → SOLID WASTE + WATERWAY / WETLAND PROTECTION → BUILDING / ROAD / UTILITY DEVELOPMENT CONTROL

HAZARD / RISK / RESILIENCE CONCEPT
- MASTER PLAN + FLOODPLAIN ZONING
- DRAIN INVENTORY + CATCHMENT / CONTOUR DESIGN

INSTITUTIONAL OR COMMUNITY RESPONSE
- SOLID WASTE + WATERWAY / WETLAND PROTECTION
- BUILDING / ROAD / UTILITY DEVELOPMENT CONTROL

CAPACITY / STATUS / EVIDENCE LIMIT
- MASTER PLAN + FLOODPLAIN ZONING
- DRAIN INVENTORY + CATCHMENT / CONTOUR DESIGN

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SOLID WASTE + WATERWAY / WETLAND PROTECTION; BUILDING / ROAD / UTILITY DEVELOPMENT CONTROL.

09

STAGE 09 SPONGE-CITY PORTFOLIO

SPONGE-CITY PORTFOLIO · CAPTURE INFILTRATE DELAY · PERMEABLE AND DETENTION SPACE · DISTRIBUTED STORAGE · MAJOR DRAIN · CONVEY RESIDUAL FLOWS · MUST REMEMBER · FLOOD RISK ANALYSIS SEPARATES RIVERINE, FLASH, PLUVIAL, COASTAL, DAM

ROOF / PLOT → capture infiltrate delay

STREET / PARK → permeable and detention space → LAKE / WETLAND → distributed storage → MAJOR DRAIN / BASIN → convey residual flows

HAZARD / RISK / RESILIENCE CONCEPT
- ROOF / PLOT → capture infiltrate delay
- STREET / PARK → permeable and detention space

INSTITUTIONAL OR COMMUNITY RESPONSE
- LAKE / WETLAND → distributed storage
- MAJOR DRAIN / BASIN → convey residual flows

CAPACITY / STATUS / EVIDENCE LIMIT
- MUST REMEMBER: Flood risk analysis separates riverine, flash, pluvial, coastal, dam-...
- ROOF / PLOT → capture infiltrate delay

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Flood risk analysis separates riverine, flash, pluvial, coastal, dam-.

10

STAGE 10 INCLUSIVE SERVICE RECOVERY LOOP

INCLUSIVE SERVICE RECOVERY LOOP · ACCESSIBLE WARNING · RELIEF + HEALTH + GRIEVANCE · SERVICE RESTORATION · HOUSING LIVELIHOOD WETLAND AND DRAIN REPAIR · SAFER PLAN AND INTER-JURISDICTION ACTION · CLOSE DISTINCTION · HEAVY RAINFALL IS NOT AUTOMATICALLY AN URBAN FLOOD; FORECAST

ACCESSIBLE WARNING → TRANSPORT → SHELTER

RELIEF + HEALTH + GRIEVANCE → SERVICE RESTORATION → HOUSING LIVELIHOOD WETLAND AND DRAIN REPAIR → FEEDBACK → SAFER PLAN AND INTER-JURISDICTION ACTION

HAZARD / RISK / RESILIENCE CONCEPT
- ACCESSIBLE WARNING → TRANSPORT → SHELTER
- RELIEF + HEALTH + GRIEVANCE → SERVICE RESTORATION

INSTITUTIONAL OR COMMUNITY RESPONSE
- HOUSING LIVELIHOOD WETLAND AND DRAIN REPAIR
- FEEDBACK → SAFER PLAN AND INTER-JURISDICTION ACTION

CAPACITY / STATUS / EVIDENCE LIMIT
- CLOSE DISTINCTION: Heavy rainfall is not automatically an urban flood; forecast rainfall...
- ACCESSIBLE WARNING → TRANSPORT → SHELTER

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Heavy rainfall is not automatically an urban flood; forecast rainfall.

11

STAGE 11 FLOOD ANSWER SPINE

FLOOD ANSWER SPINE · CLASSIFY FLOOD · TRACE CATCHMENT CITY COAST MECHANISM · MAP CWC FORECAST RESERVOIR DECISION AND LOCAL ACTION · COMBINE ZONING DRAINAGE WETLANDS STRUCTURES AND UFRMP · ADD INCLUSION COORDINATION RECOVERY AND OUTCOME QUALIFICATION · EVIDENCE LIMIT · DATE AND UNIT RAINFALL, DISCHARGE

CLASSIFY FLOOD → TRACE CATCHMENT CITY COAST MECHANISM → MAP CWC FORECAST RESERVOIR DECISION AND LOCAL ACTION → COMBINE ZONING DRAINAGE WETLANDS STRUCTURES AND UFRMP → ADD INCLUSION COORDINATION RECOVERY AND OUTCOME QUALIFICATION → EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Date and unit rainfall, discharge

HAZARD / RISK / RESILIENCE CONCEPT
- CLASSIFY FLOOD → TRACE CATCHMENT CITY COAST MECHANISM
- MAP CWC FORECAST RESERVOIR DECISION AND LOCAL ACTION

INSTITUTIONAL OR COMMUNITY RESPONSE
- COMBINE ZONING DRAINAGE WETLANDS STRUCTURES AND UFRMP
- ADD INCLUSION COORDINATION RECOVERY AND OUTCOME QUALIFICATION

CAPACITY / STATUS / EVIDENCE LIMIT
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Date and unit rainfall, discharge...
- CLASSIFY FLOOD → TRACE CATCHMENT CITY COAST MECHANISM

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Date and unit rainfall, discharge,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Date and unit rainfall, discharge,.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · STRUCTURAL MEASURES HAVE SOURCE-ACKNOWLEDGED LIMITS · INSTITUTIONAL SPLIT WITH A COORDINATION DEPENDENCY · THE URBAN · INTERNATIONAL RIVER DIMENSION · FLOODS IN UP, BIHAR, WEST · RIGOROUS TWO-FLOOD CASE SELECTION FOR 2024 Q18 · AN ADVANCED

OPTIONAL RISK / RESILIENCE DEPTH
- Structural measures have source-acknowledged limits : VisionIAS
- Institutional split with a coordination dependency : the Urban
- International river dimension : floods in UP, Bihar, West
- Rigorous two-flood case selection for 2024 Q18 : an advanced

MANDATE, WARNING OR OUTCOME LIMIT
- CAUTION: Kumbh Mela's drainage impact as an under-cited anthropogenic
- CAUTION: Lead-time-to-response-capacity mismatch : Doppler-radar-based
- Any current claim about forecasting-station density or lead-time
- CAUTION: Urban flood vulnerability is spatially concentrated in low-lying.

COMPARATIVE HAZARD / GOVERNANCE NUANCE
- CAUTION: Floodplain zoning enforcement versus urban land/housing demand
- CAUTION: Drainage retrofitting versus new-construction cost : redesigning
- NOT: The 'Urban Flooding Cell created, therefore urban floods are
- CAUTION: Investment fragmentation ("disintegrated investment decisions,"

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels